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PORTABLE ELECTRONIC DEVICE FOR STEP DETECTION AND OPERATING METHOD THEREOF

Abstract

A portable electronic device includes: a wireless communication circuit; a sensing circuit; at least one processor; and a memory operatively connected to the at least one processor and storing instructions. The at least one processor is configured to execute the instructions to cause the portable electronic device to: perform a step detection operation using the sensing circuit; receive, from a wearable device through the wireless communication circuit, first information indicating that an exercise other than walking or running is being performed; stop the step detection operation using the sensing circuit based on receiving the first information; receive, while the step detection operation is stopped and from the wearable device through the wireless communication circuit, second information indicating that the exercise ended; and resume the step detection operation based on receiving the second information.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of International Application No. PCT/KR2023/010646, filed on Jul. 24, 2023, in the Korean Intellectual Property Receiving Office, which claims priority to Korean Patent Application No. 10-2023-0011578, filed on Jan. 30, 2023, in the Korean Intellectual Property Office, and Korean Patent Application No. 10-2023-0056395, filed on Apr. 28, 2023, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND

1. Field

[0002] The disclosure relates to a portable electronic device for detecting a user's step count using an inertial sensor and a method for operating the same.

2. Description of Related Art

[0003] A portable electronic device may measure an amount of change in acceleration according to user's movement using data generated by an inertial sensor, analyze a pattern of a change in acceleration, and determine whether the analyzed pattern is a pattern corresponding to a shape of a step. The portable electronic device may detect a step when a signal characteristic value of a determined pattern satisfies a designated condition.

[0004] However, errors may occur when a portable electronic device counts steps while a user is carrying a portable electronic device and an exercise (e.g., a bicycle, a kickboard, or a motorcycle) other than walking and running is performed. The portable electronic device can resolve the errors by using location data acquired from a GPS sensor while a user is performing the exercise other than walking and running.

SUMMARY

[0005] According to an aspect of the disclosure, a portable electronic device includes: a wireless communication circuit; a sensing circuit; at least one processor connected to the wireless communication circuit and the sensing circuit; and a memory operatively connected to the at least one processor and storing instructions. The at least one processor is configured to execute the instructions to cause the portable electronic device to: perform a step detection operation using the sensing circuit; receive, from a wearable device through the wireless communication circuit, first information indicating that an exercise other than walking or running is being performed; stop the step detection operation using the sensing circuit based on receiving the first information; receive, while the step detection operation is stopped and from the wearable device through the wireless communication circuit, second information indicating that the exercise ended; and resume the step detection operation based on receiving the second information.

[0006] The at one least processor may be configured to execute the instructions to cause the

portable electronic device to: receive a first time point, from the wearable device through the wireless communication circuit, along with the first information; and subtract a number of steps counted from the first time point to a second time point from an accumulated number of steps, the second time point may be a time point at which the first information is received by the portable electronic device.

[0007] The at least one processor may be configured to execute the instructions to cause the portable electronic device to: receive, while the step detection operation is stopped and from the wearable device through the wireless communication circuit, exercise information related to a time when the exercise is performed; and determine whether to resume the step detection operation based on the exercise information.

[0008] The at least one processor may be configured to execute the instructions to cause the portable electronic device to: determine a reliability level indicating how likely the exercise is to be performed based on an exercise duration using the exercise information; resume the step detection operation based on the reliability level being greater than or equal to a reference value; and maintain stopping of the step detection operation based on the reliability level being lower than the reference value.

[0009] The at least one processor may be configured to execute the instructions to cause the portable electronic device to: determine a reliability level indicating how likely the exercise is to be performed based on an exercise duration using the exercise information; set a number of steps to start counting as a default based on the reliability level being zero; set the number of steps to start the counting as a first value greater than the default based on the determination that the reliability level is a low level; set the number of steps to start the counting as a second value greater than the first value based on the determination that the reliability level is a middle level; and maintain stopping of the step detection operation based on the determination that the reliability level is a high level.

[0010] The at least one processor may be configured to execute the instructions to cause the portable electronic device to: determine that the reliability level is zero based on the exercise duration being zero; determine the reliability level is the low level based on the exercise duration being shorter than a rest duration; determine the reliability level is the middle level based on the exercise duration being longer than the rest duration and shorter than k times the rest duration or the exercise duration being k times or longer the rest duration and shorter than N minutes; and determine the reliability level is the high level based on the exercise duration being k times or longer the rest duration and longer than or equal to N minutes.

[0011] The at least one processor may be configured to execute the instructions to cause the portable electronic device to receive, as the exercise information and from the wearable device through the wireless communication circuit, the first information including an exercise duration and a rest duration, the second information including a ratio of the exercise duration to the rest duration, or third information including a multiple value indicating how many times the exercise duration is greater than the rest duration.

[0012] According to another aspect of the disclosure, a portable electronic device includes: a touch-sensitive display; a wireless communication circuit; an input device; a sensing circuit; at least one processor connected to the touch-sensitive display, the input device, and the sensing circuit; and a memory operatively connected to the at least one processor and storing instructions. The at least one processor is configured to execute the instructions to cause the portable electronic device to: perform a step detection operation using the sensing circuit; receive, from a wearable device through the wireless communication circuit, first information indicating that an exercise other than walking or running is being performed; stop the step detection operation using the sensing circuit based on receiving the first information; receive, while the step detection operation is stopped and from the wearable device through the wireless communication circuit, second information indicating that the exercise ended; and resume the step detection operation based on receiving the

second information.

[0013] According to yet another aspect of the disclosure, a method of operating a portable electronic device includes: performing a step detection operation using a sensing circuit of the portable electronic device; receiving, from a wearable device through a wireless communication circuit of the portable electronic device, first information indicating that an exercise other than walking or running is being performed; stopping the step detection operation using the sensing circuit based on receiving of the first information; receiving, while the step detection operation is stopped and from the wearable device through the wireless communication circuit, second information indicating that the exercise is ended; and resuming the step detection operation based on receiving the second information.

[0014] The method may further include: receiving a first time point, from the wearable device through the wireless communication circuit, along with the first information; and subtracting a number of steps counted from the first time point to a second time point from an accumulated number of steps, the second time point being a time point at which the first information is received by the portable electronic device.

[0015] The method may further include: receiving, while the step detection operation is stopped and from the wearable device through the wireless communication circuit, exercise information related to a time when the exercise is performed; and determining whether to resume the step detection operation based on the exercise information.

[0016] The determining of whether to resume the step detection operation may include: determining a reliability level indicating how likely the exercise is to be performed based on an exercise duration using the exercise information; resuming the step detection operation based on the reliability level being greater than or equal to a reference value; and maintaining stopping of the step detection operation based on the reliability level being lower than the reference value.

[0017] The determining of whether to resume the step detection operation further may include: determining a reliability level indicating how likely the exercise is to be performed based on an exercise duration using the exercise information; setting a number of steps to start counting as a default based on the reliability level being zero; setting the number of steps to start the counting as a first value greater than the default based on the determination that the reliability level is a low level; setting the number of steps to start the counting as a second value greater than the first value based on the determination that the reliability level is a middle level; and maintaining stopping of the step detection operation based on the determination that the reliability level is a high level.

[0018] The determining of the reliability level may include: determining that the reliability level is zero based on the exercise duration being zero; determining the reliability level is the low level based on the exercise duration being shorter than a rest duration; determining the reliability level is the middle level based on the exercise duration being longer than the rest duration and shorter than k times the rest duration or the exercise duration being k times or longer the rest duration and shorter than N minutes; and determining the reliability level is the high level based on the exercise duration being k times or longer the rest duration and longer than or equal to N minutes.

[0019] The exercise information may include the first information including an exercise duration and a rest duration, the second information including a ratio of the exercise duration to the rest duration, or third information including a multiple value indicating how many times the exercise duration is greater than the rest duration.

[0020] According to an aspect of the present disclosure, a portable electronic device may improve the accuracy of step detection without a GPS sensor or without consuming power for driving the GPS sensor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:

[0022] FIG. 1 is a block diagram of an electronic device in a network environment according to various embodiments;

[0023] FIG. 2 illustrates a configuration of a system for detecting a step based on exercise information according to various embodiments;

[0024] FIG. 3 is a diagram illustrating a user interface (UI) screen displayed on a wearable device according to various embodiments;

[0025] FIG. 4 is a diagram illustrating a UI screen displayed on a portable electronic device according to various embodiments;

[0026] FIG. 5 is a flowchart for describing operations of a second processor for determining reliability according to various embodiments;

[0027] FIG. 6 is a flowchart for describing the operations of the second processor for controlling step detection based on reliability according to various embodiments;

[0028] FIG. 7 is a diagram for describing operations of controlling step detection based on a user's determination on a start and end of the bicycle exercise in the system of FIG. 2 according to various embodiments;

[0029] FIG. 8 is a diagram for describing the operations of controlling step detection based on the recognition of the start and end of the bicycle exercise in the system of FIG. 2 according to various embodiments;

[0030] FIG. 9 is a flowchart for describing the operations performed by the second processor of the portable electronic device based on information received from the wearable device according to various embodiments;

[0031] FIG. 10 is a flowchart for describing the operations performed by the second processor of the portable electronic device based on user input according to various embodiments;

[0032] FIG. 11 is a flowchart for describing the operations performed based on information received from the wearable device according to various embodiments; and

[0033] FIG. 12 is a flowchart for describing the operations performed based on the information received from the wearable device according to various embodiments.

DETAILED DESCRIPTION

[0034] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the drawings so that those skilled in the art to which the present disclosure pertains may easily practice the disclosure. However, the disclosure may be implemented in various different forms, and is not limited to embodiments described herein. In connection with the description of the drawings, the same or similar reference numerals may be used for the same or similar components. In addition, in the drawings and related descriptions, descriptions of well-known functions and configurations may be omitted for clarity and conciseness.

[0035] FIG. 1 is a block diagram illustrating an electronic device **101** in a network environment **100** according to various embodiments. Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a

communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0036] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0037] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0038] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0039] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0040] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the

electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0041] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0042] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0043] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0044] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0045] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0046] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0047] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0048] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0049] The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0050] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0051] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101**

and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0052] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0053] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0054] According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit

board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mm Wave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0055] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0056] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0057] FIG. 2 illustrates a configuration of a system **200** for detecting a step based on exercise information according to one or more embodiments. Referring to FIG. 2, the system **200** may include a wearable device **210** and a portable electronic device **220**.

[0058] The wearable device **210** (e.g., a smart watch) may include a first communication circuit **211**, a first sensing circuit **212**, a first display **213**, a first input module **214**, a first memory **218**, and a first processor **219**. The above-described components of the wearable device **210** may be operatively or electrically connected to each other. The first communication circuit **211**, the first sensing circuit **212**, the first display **213**, the first input module **214**, the first memory **218**, and the first processor **219** are implemented to be substantially the same as the wireless communication module **192**, the sensor module **176**, the display module **160**, the input module **150**, the memory **130**, and the processor **120** of FIG. 1, respectively, and thus, may perform the same functions.

[0059] The first communication circuit **211** may include a short-range wireless communication circuit configured to establish a wireless communication channel **201** between the wearable device **210** and the portable electronic device **220** and perform data communication between the two devices **210** and **220** through the established wireless communication channel **201**. For example, the first communication circuit **211** may include a Bluetooth communication circuit. Additionally, the first communication circuit **211** may include a Wi-Fi communication circuit. The first processor **219** may perform data communication with the portable electronic device **220** through the short-range wireless communication channel (e.g., a Bluetooth communication channel or a Wi-Fi

communication channel) established by the first communication circuit **211**.

[0060] The first sensing circuit **212** is configured to recognize a bicycle exercise and generate data used to monitor the process of the bicycle exercise, and may provide the generated data to the first processor **219**. For example, the first sensing circuit **212** may include an inertia measurement unit (IMU) that is configured to include an acceleration sensor, a gyroscope, and a magnetometer. Additionally, or alternatively, the first sensing circuit **212** is configured to recognize a user's step during a step detection operation, generate data used to monitor the process of the user's walk or run, and may provide the generated data to the first processor **219**.

[0061] The bicycle exercise algorithm **230** is composed of instructions, and may be stored in the first memory **218** and executed by the first processor **219**. The first processor **219** may store data generated by executing the bicycle exercise algorithm **230** in the first memory **218**.

[0062] The bicycle exercise algorithm **230** may be configured to identify whether the bicycle is in operation (in other words, while a user is performing the bicycle exercise) using data received from the first sensing circuit **212**. For example, the bicycle exercise algorithm **230** may monitor physical quantities (e.g., acceleration, angular velocity, magnetic field) from the data received from the first sensing circuit **212**. When the bicycle is in operation, the physical quantities acquired through monitoring may satisfy predetermined bicycle exercise patterns. When the bicycle is stopped, the acquired physical quantities may not satisfy the bicycle exercise patterns. The bicycle exercise algorithm **230** may recognize that the user is performing the bicycle exercise when the acquired physical quantities satisfy the bicycle exercise patterns.

[0063] According to one or more embodiments, the bicycle exercise algorithm **230** may be configured to transmit, to the portable electronic device **220** through the first communication circuit **211**, a first notification message that notifies the portable electronic device **220** that the bicycle exercise is started based on recognizing that the bicycle exercise is started. For example, the bicycle exercise algorithm **230** may recognize that the bicycle exercise is started when the acquired physical quantities satisfy the bicycle exercise patterns, and transmit the first notification message to the portable electronic device **220** through the first communication circuit **211** accordingly. For another example, the bicycle exercise algorithm **230** may recognize that the bicycle exercise is started when the physical quantities satisfying the bicycle exercise patterns are repeatedly acquired for a designated time (e.g., 3 minutes). Based on recognizing that the bicycle exercise is started, the bicycle exercise algorithm **230** may transmit the first notification message to the portable electronic device **220** through the first communication circuit **211**. The bicycle exercise algorithm **230** may include information indicating the time point at which the bicycle exercise patterns is first satisfied in a first notification message and transmit the first notification message to the portable electronic device **220**. The bicycle exercise algorithm **230** may stop transmitting the scheduled first notification message based on the physical quantity that do not satisfy the bicycle exercise patterns being acquired before the designated time has elapsed. In the following description, the time point at which the bicycle exercise patterns are first satisfied is referred to as a 'first time point', and the time point at which the first notification message is transmitted after the first time point is referred to as a 'second time point'.

[0064] According to one or more embodiments, the bicycle exercise algorithm **230** may be configured to repeatedly transmit, to the portable electronic device **220** through the first communication circuit **211**, every designated time a second notification message that enables the portable electronic device **220** to recognize that the bicycle exercise is being maintained while the bicycle exercise is being maintained after the first notification message is transmitted.

[0065] According to one or more embodiments, the bicycle exercise algorithm **230** may be configured to monitor the process of the bicycle exercise using the data received from the first sensing circuit **212** and record the monitored result in the first memory **218**. For example, the bicycle exercise algorithm **230** may measure the time (hereinafter, exercise duration) when the physical quantities acquired through the monitoring are recognized as satisfying the bicycle

exercise patterns and the time (hereinafter, rest duration) when the physical quantities are recognized as not satisfying the bicycle exercise patterns based on the data received from the first sensing circuit **212**, and store the measured results in the first memory **218**.

[0066] According to one or more embodiments, the bicycle exercise algorithm **230** may be configured to repeatedly transmit bicycle exercise information related to the exercise duration during a predetermined duration of time to the portable electronic device **220** through the first communication circuit **211**. For example, the bicycle exercise algorithm **230** may measure the exercise duration and the rest duration in the designated unit time (e.g., 10 minutes), and provide the bicycle exercise information including the measured exercise duration and rest duration to the portable electronic device **220** every time the unit time elapses. For another example, the bicycle exercise algorithm **230** may calculate a ratio of the exercise duration to the rest duration during the designated unit time, and provide bicycle exercise information including the calculated ratio to the portable electronic device **220** every time the unit time elapses. For another example, the bicycle exercise algorithm **230** may calculate a multiple value that indicates how many times the exercise duration is greater than the rest duration during the designated unit time, and provide the bicycle exercise information including the calculated multiple value to the portable electronic device **220** every time the unit time elapses.

[0067] For example, a transmission time point of the bicycle exercise information may be substantially the same as that of the second notification message. The bicycle exercise algorithm **230** may include the bicycle exercise information in the second notification message and provide the second notification message to the portable electronic device **220**.

[0068] According to one or more embodiments, the bicycle exercise algorithm **230** may be configured to recognize that the bicycle exercise is ended based on the rest duration during which the bicycle is recognized as not being operated exceeding a designated threshold, and end the monitoring and recording accordingly. In addition, the bicycle exercise algorithm **230** may be configured to transmit, to the portable electronic device **220** through the first communication circuit **211**, a third notification message that notifies the portable electronic device **220** that the bicycle exercise is ended based on the rest duration exceeding the specified threshold.

[0069] The first communication circuit **211** may further include, for example, a GNSS communication circuit as a communication circuit for identifying a geographical location of the wearable device **210**. The bicycle exercise algorithm **230** may additionally use data acquired from the GNSS communication circuit in addition to data acquired from the first sensing circuit **212** in performing the recognition and monitoring of bicycle exercise.

[0070] The first sensing circuit **212** may further include a heart rate detection sensor. The bicycle exercise algorithm **230** may further use data received from the heart rate detection sensor to perform the recognition and monitoring of the bicycle exercise. For example, the bicycle exercise algorithm **230** may acquire vibration patterns (e.g., vibration frequency, amplitude, intensity) from data received from an inertial sensor, acquire a speed using data received from the GNSS communication circuit, and acquire a heart rate from data received from a heart rate detection sensor. The bicycle exercise algorithm **230** may recognize that the user is performing the bicycle exercise based on whether the acquired information satisfies the predetermined bicycle exercise patterns.

[0071] According to one or more embodiments, the first display **213** and/or the first input module **214** may be used as input devices that notify the portable electronic device **220** of the start and end of the bicycle exercise.

[0072] FIG. **3** is a diagram illustrating an example of a user interface (UI) screen displayed on the wearable device **210**. Referring to FIG. **3**, the first processor **219** may provide, to a user through the first display **213**, a first UI screen **310** that enables the user to determine the start of the bicycle exercise. The bicycle exercise algorithm **230** may receive a user input indicating the start of the bicycle exercise through the first UI screen **310** displayed on the first display **213**. For example,

when a user touches a display object **311** (e.g., an icon) that is displayed as ‘cycling’ on the first UI screen **310**, the bicycle exercise algorithm **230** may recognize that a user input corresponding to a touch is transmitted from the first display **213** to the first processor **219**.

[0073] Referring to FIG. 2, the first processor **219** may provide, to a user through the first display **213**, a second UI screen (not illustrated) that enables the user to determine the end of the bicycle exercise. The bicycle exercise algorithm **230** may receive a user input indicating the end of the bicycle exercise through the second UI screen displayed on the first display **213**.

[0074] The bicycle exercise algorithm **230** may also receive the user input indicating the start or end of the bicycle exercise through the first input module **214** (e.g., a microphone or a physical button).

[0075] According to one or more embodiments, the bicycle exercise algorithm **230** may be configured to transmit the first notification message to the portable electronic device **220** based on the reception of the user input indicating the start of the bicycle exercise. The bicycle exercise algorithm **230** may be configured to transmit the third notification message to the portable electronic device **220** based on the reception of the user input indicating the end of the bicycle exercise.

[0076] According to one or more embodiments, the wearable device **210** may include various exercise algorithms configured to perform substantially the same function as the bicycle exercise algorithm **230** described above, except that the type of exercises is different. For example, the wearable device may include an exercise algorithm configured to identify whether the user is riding a kickboard or a motorcycle by using the data received from the sensing circuit (e.g., the first sensing circuit **212** of FIG. 2). In addition, the wearable device **210** may provide the UI screen to the user through the first display **213** that may allow the user to determine the start and end of the kickboard or motorcycle exercise.

[0077] The portable electronic device **220** (e.g., a smart phone) may include a second communication circuit **221**, a second sensing circuit **222**, a second display **223**, a second input module **224**, a second memory **228**, and a second processor **229**. The above-described components of the portable electronic device **220** may be operatively or electrically connected to each other. The second communication circuit **221**, the second sensing circuit **222**, the second display **223**, the second input module **224**, the second memory **228**, and the second processor **229** are implemented to be substantially the same as the wireless communication module **192**, the sensor module **176**, the display module **160**, the input module **150**, the memory **130**, and the processor **120** of FIG. 1, respectively, and thus, may perform the same functions.

[0078] The second communication circuit **221** may include the short-range wireless communication circuit configured to establish the wireless communication channel **201** between the wearable device **210** and the portable electronic device **220** and perform data communication between the two devices **210** and **220** through the established wireless communication channel **201**. For example, the second communication circuit **221** may include a Bluetooth communication circuit. Additionally, the second communication circuit **221** may include a WiFi communication circuit. The second processor **229** may perform data communication with the wearable device **210** through the short-range wireless communication channel (e.g., a Bluetooth communication channel or a WiFi communication channel) established by the second communication circuit **221**.

[0079] The second sensing circuit **222** may be configured to generate data used to detect a step and provide the generated data to the second processor **229**. For example, the second sensing circuit **222** may include the inertia measurement unit (IMU) that is configured to include the acceleration sensor, the gyroscope, and the magnetometer.

[0080] A step detection module **240**, a detection control module **250**, a step number correction module **260**, a reliability determination module **270**, and a condition configuration module **280** may be configured as instructions, stored in a second memory **228**, and executed by a second processor **229**. The second processor **229** may store information indicating step information (e.g., the number

of steps, distance, walking time zone) measured using the above-described modules **240** to **280** in the second memory **228**.

[0081] The step detection module **240** may be configured to detect a step based on data received from the second sensing circuit **222**. For example, the step detection module **240** may detect vibration patterns (e.g., vibration frequency, amplitude, and intensity) that satisfy predetermined step patterns from the data received from the first sensing circuit **212**. The step detection module **240** may count steps whenever the vibration patterns are detected and store the counted number of steps in the second memory **228**.

[0082] The detection control module **250** may be configured to control the step detection module **240**. The step detection module **240** may perform or stop the step detection operation under the control of the detection control module **250**. The step detection module **240** may continue the step detection but stop the counting operation based on the control of the detection control module **250**. For example, the step detection module **240** may start counting from an N+1 step without counting steps detected from first to Nth.

[0083] According to one or more embodiments, the detection control module **250** may control the step detection module **240** to stop the step detection operation based on the reception of the first notification message (e.g., a message notifying that an exercise (bicycle, kick board, or motorcycle) other than walking and running is started) from the wearable device **210** through the second communication circuit **221**. After the stop, the detection control module **250** may receive the third notification message (e.g., a message notifying that an exercise is ended) from the wearable device **210** through the second communication circuit **221**. The detection control module **250** may control the step detection module **240** to resume the step detection operation based on the reception of the third notification message.

[0084] According to one or more embodiments, the second display **223** and/or the second input module **224** may be used as input devices that enables the user to determine the start and end of the bicycle exercise (or, kick board or motorcycle exercise).

[0085] FIG. 4 illustrates an example of a UI screen displayed on the portable electronic device **220**. Referring to FIG. 4, the second processor **229** may provide, to a user through the second display **223**, a third UI screen **410** that enables the user to determine the start of the bicycle exercise. The detection exercise algorithm **250** may receive a user input indicating the start of the bicycle exercise through the third UI screen **410** displayed on the second display **223**. For example, the detection control module **250** may receive a user's touch input for a display object **411** (e.g., an icon) indicated as 'start' on the third UI screen **410** from the second display **223**. The second processor **229** may transmit a message indicating the start of the bicycle exercise to the wearable device **210** through the second communication circuit **221** in response to the reception of the user input indicating the start of the bicycle exercise through the second display **223**. The bicycle exercise algorithm **230** may recognize the start of the bicycle exercise through the message received by the wearable device **210**. The second processor **229** may not transmit the start message to the wearable device.

[0086] Referring to FIG. 2, the second processor **229** may provide, to a user through the second display **223**, a fourth UI screen (not illustrated) that enables the user to determine the end of the bicycle exercise. The detection control module **250** may receive a user input indicating the end of the bicycle exercise through the fourth UI screen displayed on the second display **223**. The second processor **229** may transmit a message indicating the end of the bicycle exercise to the wearable device **210** through the second communication circuit **221** in response to the reception of the user input indicating the end of the bicycle exercise through the second display **223**. The bicycle exercise algorithm **230** may recognize the end of the bicycle exercise through the message received by the wearable device **210**. The second processor **229** may not transmit the end message to the wearable device.

[0087] According to one or more embodiments, the portable electronic device **220** may provide, to

a user through the second display **223**, a UI screen that enables the user to determine the start and end of a kick board or motorcycle exercise.

[0088] The detection control module **250** may also receive a user input (e.g., a user input indicating the start of the bicycle, the kick board, or the motorcycle exercise) indicating the start of the exercise and a user input (e.g., a user input indicating the end of the bicycle, the kick board, or the motorcycle exercise) indicating the end of the exercise through the second input module **224** (e.g., a microphone or a physical button).

[0089] The detection control module **250** may control the step detection module **240** to stop the step detection operation based on the reception of the user input indicating the start of the exercise. The detection control module **250** may control the step detection module **240** to resume the step detection operation based on the reception of the user input indicating the end of the exercise.

[0090] As described above, the detection control module **250** may receive the user input from the second display **223** or the second input module **224** as an event that triggers stopping or resuming the step detection operation. The detection control module **250** may receive the notification message from the wearable device **210** as the event that triggers stopping or resuming the step detection operation. According to one or more embodiments, the detection control module **250** may maintain the stop of the step detection without resuming the step detection when the notification message to resume the step detection is received from the wearable device **210** while the step detection operation is stopped by the user input. According to one or more embodiments, the detection control module **250** may control the step detection module **240** to resume the step detection when the user input to resume the step detection is received from the second display **223** or the second input module **224** while the step detection operation is stopped by the notification message.

[0091] The step number correction module **260** may be configured to correct the number of steps (e.g., the accumulated number of steps measured and stored during a day) counted by the step detection module **240** and stored in the second memory **228** based on the exercise information (e.g., exercise information on the bicycle, the kick board, or the motorcycle) received from the wearable device **210** through the second communication circuit **221**.

[0092] According to one or more embodiments, the wearable device **210** may include information indicating a first time point at which the exercise patterns (e.g., exercise patterns regarding the bicycle, the kick board, or the motorcycle) are first satisfied in the first notification message and transmit the first notification message to the portable electronic device **220**. The step number correction module **260** may confirm the first time point in the first notification message. The step number correction module **260** may correct the number of steps by subtracting the number of steps counted by the step detection module **240** from the first time point to the second time point at which the first notification message is received from the number of steps stored in the second memory **228**.

[0093] The reliability determination module **270** may be configured to receive the exercise information (e.g., exercise information on the bicycle, the kick board, or the motorcycle) from the wearable device **210**. The reliability determination module **270** may be configured to determine reliability level (a “reliability”) that indicates how likely the corresponding exercise is to be performed from the received exercise information. The reliability determination module **270** may be configured to determine the reliability as a high level in proportion to the exercise duration.

[0094] According to one or more embodiments, the reliability determination module **270** may receive the exercise information related to the exercise duration during a predetermined duration of time from the wearable device **210** through the second communication circuit **221**. For example, the reliability determination module **270** may receive, from the wearable device **210**, the first information including the exercise duration and the rest duration, the second information including a ratio of the exercise duration to the rest duration, or the third information including a multiple value that indicates how many times the exercise duration is greater than the rest duration. Based

on the received information, the reliability determination module **270** may determine the reliability as one of the no reliability, low level, the middle level, and the high level. The no reliability may be a reliability level of zero, the low level may be a reliability level above zero, the middle level may be a reliability level above the no reliability level and the low reliability level, and the high level may be a reliability level above the no reliability level, the low reliability level, and the middle reliability level.

[0095] In the following embodiments, each operation may be performed sequentially, but is not necessarily performed sequentially. For example, the order of each operation may be changed, and at least two operations may be performed in parallel.

[0096] FIG. **5** is a flowchart for describing operations of a second processor **229** for determining reliability according to one or more embodiments. When the reliability determination module **270** is executed, the second processor **229** may perform the following operations based on the information (the first information, the second information, or the third information) received from the wearable device **210**. In the description of FIG. **5**, the exercise is specified as ‘bicycle exercise’, but is not limited thereto. That is, the operations of FIG. **5** may also be applied to other exercises (e.g., the kick board or the motorcycle).

[0097] It may be understood that operations **510** to **570** are performed by the second processor (e.g., the second processor **229** of FIG. **2**) of the portable electronic device (e.g., the portable electronic device **220** of FIG. **2**).

[0098] In operation **510**, the second processor **229** may identify whether there is the exercise duration from the received information. If there is no exercise duration (exercise duration=0 seconds), in operation **511**, the second processor **229** may determine “no reliability (the user is unlikely to be riding a bicycle)”.

[0099] When it is determined that there is the exercise duration, the second processor **229** may identify whether the exercise duration is longer than the rest duration from the received information in operation **520**. When the exercise duration is not longer than the rest duration, in operation **530**, the second processor **229** may determine the reliability as the low level.

[0100] When the exercise duration is longer than the rest duration, in operation **540**, the second processor **229** may identify whether the exercise duration is k times (e.g., k is a number greater than 2) or longer the rest duration from the received information.

[0101] When the exercise duration is k times or longer the rest duration, in operation **550**, the second processor **229** may identify whether the exercise duration is longer than or equal to a certain time (e.g., N minutes) from the received information.

[0102] When the exercise duration is shorter than k times the rest duration or the exercise duration is shorter than the certain time, in operation **560**, the second processor **229** may determine the reliability as the middle level.

[0103] When the exercise duration is k times or more the rest duration or the exercise duration is longer than or equal to the certain time, in operation **570**, the second processor **229** may determine the reliability as the high level.

[0104] The technical idea of the present disclosure is not limited to classifying reliability into four levels (no reliability, low, middle, high) as described above, and may be classified into five levels or less or five levels or more.

[0105] According to one or more embodiments, the detection control module **250** may control the step detection module **240** based on the reliability determined by the reliability determination module **270**. The detection control module **250** may control the step detection module **240** to stop the step detection operation when the determined reliability is greater than or equal to a certain level (e.g., the high level of FIG. **5**). The detection control module **250** may control the step detection module **240** to perform the step detection operation when the determined reliability is below a certain level (e.g., low level, middle level, or no reliability of FIG. **5**).

[0106] The condition configuration module **280** may be configured to set a condition for starting

the operation (hereinafter, “count”) of counting steps based on the reliability determined by the reliability determination module **270**. Here, the condition may include the number of steps or the characteristic values (e.g., the vibration frequency, the amplitude, the intensity) of the step pattern. [0107] According to one or more embodiments, the condition configuration module **280** may control the step detection module **240** to start counting from an Nth step based on the determination that there is no reliability. For example, the step detection module **240** may count steps from an Nth step after a user takes his/her first step. The condition configuration module **280** may control the step detection module **240** to start the counting from an N+ath step (a is a positive integer) based on the determination that the reliability is the low level. The condition configuration module **280** may control the step detection module **240** to start the counting from an N+bth step (b is an integer greater than a) based on the determination that the reliability is the middle level.

[0108] FIG. **6** is a flowchart for describing the operations of the second processor **229** for controlling step detection based on the reliability according to one or more embodiments. The detection control module **250** and the condition configuration module **280**, may cause, when executed, the second processor **229** to perform the following operations based on the reliability determined by the reliability determination module **270**.

[0109] It may be understood that operations **610** to **650** are performed by the second processor (e.g., the second processor **229** of FIG. **2**) of the portable electronic device (e.g., the portable electronic device **220** of FIG. **2**).

[0110] In operation **610**, the second processor **229** may identify whether it has been determined that there is the reliability. When it is determined that there is no reliability, in operation **620**, the second processor **229** may set the number of steps to start the counting as a default (e.g., N). The second processor **229** may start the counting from the step in the order corresponding to the value set to start the counting. For example, when the default is set to 5, the second processor **229** may start the counting from the fifth detected step without counting from the first detected step to the fourth detected step.

[0111] When it is determined that there is the reliability, in operation **630**, the second processor **229** may identify whether the determined reliability is greater than or equal to a reference value. When the determined reliability is less than the reference value, in operation **640**, the second processor **229** may change the number of steps to start the counting to a value greater than the default. When the determined reliability is greater than or equal to the reference value, in operation **650**, the second processor **229** may stop the step detection operation using the second sensing circuit **222**. For example, the reliability may be determined as the low level, the middle level, or the high level. When it is determined that the reliability is the low level, the second processor **229** may set a first value (e.g., N+a) greater than the default as a value to start the counting. When it is determined that the reliability is the middle level, the second processor **229** may set a second value (e.g., N+b) greater than the first value as a value to start the counting. When it is determined that the reliability is the high level, the second processor **229** may stop the step detection.

[0112] FIG. **7** is a diagram for describing operations of controlling step detection based on the user's determination on the start and end of the bicycle exercise in the system **200** of FIG. **2** according to one or more embodiments. In FIG. **7**, a health application **701** (e.g., application **146** of FIG. **1**) is configured to include a detection control module (e.g., the detection control module **250** of FIG. **2**), and may be stored in a second memory (e.g., the second memory **228** of FIG. **2**) of the portable electronic device **220** and executed by a second processor (e.g., the second processor **229** of FIG. **2**). In the description of FIG. **7**, the exercise is specified as ‘bicycle exercise’, but is not limited thereto. That is, the operations described with reference to FIG. **7** may also be applied to other exercises (e.g., the kick board or the motorcycle).

[0113] In the wearable device **210**, a first processor (e.g., the first processor **219** of FIG. **2**) may receive a first user input **711** indicating the start of the bicycle exercise from a user through a first display (e.g., the first display **213** of FIG. **2**) or a first input module (e.g., the first input module **214**

of FIG. 2). In response to receiving the first user input **711**, the first processor **219** may transmit a start notification message **721** to the portable electronic device **220** through a first communication circuit (e.g., the first communication circuit **211** of FIG. 2). In the portable electronic device **220**, the second processor **229** may receive the start notification message **721** through a second communication circuit (e.g., the second communication circuit **221** of FIG. 2). The health application **701** may transmit the start notification message **721** received by the portable electronic device **220** to a step detection module (e.g., the step detection module **240** of FIG. 2). The step detection module **240** may stop the step detection based on the reception of the start notification message **721**. The health application **701** may store start time point information indicating that the bicycle exercise is started at the time point at which the start notification message **721** is received in the second memory **228**.

[0114] In the wearable device **210**, the first processor **219** may transmit, to the portable electronic device **220** through the first communication circuit **211**, an exercise notification message **722** that notifies the portable electronic device **220** that the bicycle exercise is performing whenever a given period of time (N minutes) elapses after notifying the portable electronic device **220** of the start of the bicycle exercise. The health application **701** may transmit the exercise notification message **722** received by the portable electronic device **220** to the step detection module **240**. The step detection module **240** may maintain the stop of the step detection based on the reception of the exercise notification message **722**. The exercise notification message **722** may not be transmitted to the portable electronic device **220** even after a given period of time has elapsed due to the wireless connection being disconnected between the two devices **210** and **220**. In this case, the step detection module **240** may resume the step detection.

[0115] Before a certain time (N minutes) has elapsed after the exercise notification message **722** is transmitted to the portable electronic device **220**, the first processor **219** may receive the second user input **712** indicating the end of the bicycle exercise from the user through the first display **213** or the first input module **214**. In response to receiving the second user input **712**, the first processor **219** may transmit an end notification message **723** to the portable electronic device **220** through the first communication circuit **211**. In the portable electronic device **220**, the second processor **229** may receive the end notification message **723** through the second communication circuit **221**. The health application **701** may transmit the end notification message **723** received by the portable electronic device **220** to the step detection module **240**. The step detection module **240** may resume the step detection based on the reception of the end notification message **723**. The health application **701** may store the end time point information indicating that the bicycle exercise is ended at the time point at which the end notification message **723** is received in the second memory **228**.

[0116] FIG. 8 is a diagram for describing the operations of controlling step detection based on the recognition of the start and end of the bicycle exercise in the system **200** of FIG. 2 according to one or more embodiments. In FIG. 8, the health application **701** is configured to include a detection control module (e.g., the detection control module **250** of FIG. 2), a step number correction module (e.g., the step number correction module **260** of FIG. 2), a reliability determination module (e.g., the reliability determination module **270** of FIG. 2), and a condition configuration module (e.g., the condition configuration module **280** of FIG. 2), and may be stored in a second memory (e.g., the second memory **228** of FIG. 2) of the portable electronic device **220** and executed by a second processor (e.g., the second processor **229** of FIG. 2). In the description of FIG. 8, the exercise is specified as 'bicycle exercise', but is not limited thereto. That is, the operations described with reference to FIG. 8 may also be applied to other exercises (e.g., the kick board or the motorcycle).

[0117] In the wearable device **210**, the first processor (e.g., the first processor **219** of FIG. 2) may recognize that physical quantities acquired through a first sensing circuit (e.g., the first sensing circuit **212** of FIG. 2) satisfy bicycle exercise patterns at the first time point A. Based on the fact that the physical quantities satisfying the bicycle exercise patterns are continuously acquired for a

designated time **801** from a first time point A, the first processor **219** may transmit a start notification message **821** to the portable electronic device **220** through the first communication circuit (e.g., the first communication circuit **211** of FIG. 2) at a second time point B. The first processor **219** may include the start time point information on the first time point A in the start notification message **821** and transmit the start notification message **821** to the portable electronic device **220**. In the portable electronic device **220**, the second processor **229** may receive the start notification message **821** through the second communication circuit (e.g., the second communication circuit **221** of FIG. 2) at the second time point B. The health application **701** may transmit the start notification message **821** received by the portable electronic device **220** to the step detection module (e.g., the step detection module **240** of FIG. 2). The step detection module **240** may stop the step detection based on the reception of the start notification message **821**. The health application **701** may identify the first time point A in the start notification message **821**, and the step detection module **240** may correct the number of steps by subtracting the number of steps counted from the first time point A to the second time point B from the number of steps (e.g., the number of daily steps) stored in the second memory **228**. The health application **701** may store the start time point information indicating that the bicycle exercise is started at the first time point A in the second memory **228**.

[0118] In the wearable device **210**, the first processor **219** may transmit, to the portable electronic device **220** through the first communication circuit **211**, an exercise notification message **822** that notifies the portable electronic device **220** that the bicycle exercise is performing whenever the given period of time (N minutes) elapses after notifying the portable electronic device **220** of the start of the bicycle exercise. The first processor **219** may include the exercise duration information in the start notification message **821** and transmit the start notification message **821** to the portable electronic device **220**. For example, the first processor **219** may include the first information indicating the exercise duration and the rest duration during a predetermined duration of time (N minutes), the second information indicating the ratio of the exercise duration to the rest duration, or the information that indicates how many times the exercise duration is greater than the rest duration in the exercise notification message **822**, and transmit the exercise notification message **822** to the portable electronic device **220**.

[0119] The health application **701** may determine whether to maintain the stop of the step detection based on the exercise duration information included in the exercise notification message **822**. According to one or more embodiments, the health application **701** may determine the reliability of the bicycle exercise as one of no reliability, the low level, the middle level, and the high level, as exemplarily described with reference to FIG. 5, based on the exercise duration information included in the exercise notification message **822**. When the determined reliability is the high level, the health application **701** may transmit the exercise notification message **822** to the step detection module **240** to maintain the stop of the step detection. When the determined reliability is the low level or the middle level, the health application **701** may control the step detection module **240** to change the number of steps to start the counting to the value greater than the default (e.g., operation **640**) and resume the step detection.

[0120] The exercise notification message **822** may not be transmitted to the portable electronic device **220** even after the given period of time has elapsed due to the wireless connection being disconnected between the two devices **210** and **220**. In this case, the step detection module **240** may resume the step detection.

[0121] Before a certain time (N minutes) has elapsed after the exercise notification message **822** is transmitted to the portable electronic device **220**, the first processor **219** in the wearable device **210** may recognize that the physical quantities acquired through the first sensing circuit **212** do not satisfy the bicycle exercise patterns at the third time point C. Based on the fact that the physical quantities not satisfying the bicycle exercise patterns are continuously acquired for the designated time **802** from the third time point C, the first processor **219** may transmit an end notification

message **823** to the portable electronic device **220** through the first communication circuit **211** at a fourth time point D. The first processor **219** may include the end time point information on the third time point C in the end notification message **823** and transmit the end notification message **823** to the portable electronic device **220**. The second processor **229** in the portable electronic device **220** may receive the end notification message **823** through the second communication circuit **221** at the fourth time point D. The health application **701** may transmit the end notification message **823** received by the portable electronic device **220** to the step detection module **240**. The step detection module **240** may resume the step detection based on the reception of the end notification message **823**. The health application **701** may identify the third time point C in the end notification message **823** and store the end time point information indicating that the bicycle exercise is ended at the third time point C in the second memory **228**. FIG. **9** is a flowchart for describing operations performed by the second processor **229** of the portable electronic device **220** based on the information received from the wearable device **210** according to one or more embodiments. In the description of FIG. **9**, the exercise is specified as 'bicycle exercise', but is not limited thereto. That is, the operations described with reference to FIG. **9** may also be applied to other exercises (e.g., the kick board or the motorcycle).

[0122] It may be understood that operations **910** to **940** are performed by the second processor (e.g., the second processor **229** of FIG. **2**) of the portable electronic device (e.g., the portable electronic device **220** of FIG. **2**).

[0123] In operation **910**, the second processor **229** may receive the first information (e.g., the start notification message **721** or **821**) that indicates that the bicycle exercise is performing from the wearable device **210** through the second communication circuit **221**.

[0124] In operation **920**, the second processor **229** may stop the step detection operation using the second sensing circuit **222** based on the reception of the first information.

[0125] While the step detection operation is stopped, in operation **930**, the second processor **229** may receive the second information (e.g., the end notification message **723** or **823**) that indicates that the bicycle exercise is ended from the wearable device **210** through the second communication circuit **221**.

[0126] In operation **940**, the second processor **229** may resume the step detection based on the reception of the second information.

[0127] FIG. **10** is a flowchart for describing the operations performed by the second processor **229** of the portable electronic device **220** based on the user input according to one or more embodiments. In the description of FIG. **10**, the exercise is specified as 'bicycle exercise', but is not limited thereto. That is, the operations described with reference to FIG. **10** may also be applied to other exercises (e.g., the kick board or the motorcycle).

[0128] It may be understood that operations **1010** to **1040** are performed by the second processor (e.g., the second processor **229** of FIG. **2**) of the portable electronic device (e.g., the portable electronic device **220** of FIG. **2**).

[0129] In operation **1010**, the second processor **229** may receive the first user input indicating the start of the bicycle exercise from the second display **223** or the second input module **224**.

[0130] In operation **1020**, the second processor **229** may stop the step detection based on the reception of the first user input.

[0131] While the step detection operation is stopped, in operation **1030**, the second processor **229** may receive the second user input indicating the end of the bicycle exercise from the second display **223** or the second input module **224**.

[0132] In operation **1040**, the second processor **229** may resume the step detection based on the reception of the second user input.

[0133] FIG. **11** is a flowchart for describing the operations performed based on the information received from the wearable device **210** according to one or more embodiments. In the description of FIG. **11**, the exercise is specified as 'bicycle exercise', but is not limited thereto. That is, the

operations described with reference to FIG. 11 may also be applied to other exercises (e.g., the kick board or the motorcycle).

[0134] It may be understood that operations **1110** to **1130** are performed by the second processor (e.g., the second processor **229** of FIG. 2) of the portable electronic device (e.g., the portable electronic device **220** of FIG. 2).

[0135] In operation **1110**, the second processor **229** may receive the start notification message (e.g., the start notification message **821**) indicating the start of the bicycle exercise from the wearable device **210** through the second communication circuit **221**.

[0136] In operation **1120**, the second processor **229** may identify the information indicating the first time point at which the bicycle exercise patterns are first satisfied (e.g., as the time point A in FIG. 8, the time point at which the physical quantities acquired from the wearable device **210** are first recognized to satisfy the bicycle exercise patterns) in the start notification message.

[0137] In operation **1130**, the second processor **229** may correct the accumulated number of steps by subtracting the number of steps counted from the first time point to the second time point at which the start notification message is received from the accumulated number of steps (e.g., the number of daily steps) stored in the second memory **228**.

[0138] FIG. 12 is a flowchart for describing the operations performed based on the information received from the wearable device **210** according to one or more embodiments. In the description of FIG. 12, the exercise is specified as 'bicycle exercise', but is not limited thereto. That is, the operations described with reference to FIG. 12 may also be applied to other exercises (e.g., the kick board or the motorcycle).

[0139] It may be understood that operations **1210** to **1230** are performed by the second processor (e.g., the second processor **229** of FIG. 2) of the portable electronic device (e.g., the portable electronic device **220** of FIG. 2).

[0140] While the step detection is stopped, in operation **1210**, the second processor **229** may receive the bicycle exercise information (e.g., exercise notification message **722** or **822**) related to the time when the bicycle exercise was performed during a predetermined duration of time from the wearable device **210** through the second communication circuit **221**.

[0141] In operation **1220**, the second processor **229** may determine the reliability based on the exercise duration recognized using the bicycle exercise information. For example, as described with reference to FIG. 5, the second processor **229** may determine the reliability as one of "no", the low level, the middle level, and the high level based on the exercise duration.

[0142] In operation **1230**, the second processor **229** may determine whether to resume the step detection based on the determined reliability. For example, when it is determined that the reliability is the high level, the second processor **229** may maintain the stop of the step detection. When it is determined that there is no reliability or when it is determined that the reliability is the middle level or the low level, the second processor **229** may resume the step detection. When it is determined that there is no reliability, the second processor **229** may set the number of steps to start the counting as the default (N numbers). When it is determined that the reliability is the middle level or the low level, the second processor **229** may set the number of steps to start the counting as the value greater than the default. For example, when it is determined that the reliability is the low level, the second processor **229** may set the first value (e.g., N+a) greater than the default (N numbers) as the value to start the counting. When it is determined that the reliability is the middle level, the second processor **229** may set the second value (e.g., N+b) greater than the first value as the value to start the counting.

[0143] In the above description, the prefixes "first," "second," "third," etc., are only used to distinguish between the same names and do not have any special meaning in themselves, such as importance or order.

[0144] According to one or more embodiments, the portable electronic device (e.g., portable electronic device **220**) includes: a wireless communication circuit; a sensing circuit; a processor

connected to the wireless communication circuit and the sensing circuit; and a memory operatively connected to the processor. The memory may store instructions which cause, when executed, the processor to receive first information indicating that an exercise other than walking and running is being performed, from a wearable device (e.g., wearable device **210**) through the wireless communication circuit. The instructions may cause the processor to stop the step detection operation using the sensing circuit based on the reception of the first information. The instructions may cause the processor to receive the second information indicating that the exercise is ended, from the wearable device through the wireless communication circuit, while the detection operation is stopped. The instructions may cause the processor to resume the detection operation based on the reception of the second information.

[0145] The information indicating the first time point along with the first information may be received by the portable electronic device. The instructions may cause the processor to subtract the number of steps counted from the first time point to the second time point at which the first information is received from the accumulated number of steps.

[0146] The instructions may cause the processor to receive the exercise information related to the time when an exercise is performed during a predetermined duration of time from the wearable device through the wireless communication circuit, while the detection operation is stopped. The instructions may cause the processor to determine whether to resume the detection operation based on the exercise information.

[0147] The instructions may cause the processor to determine the reliability that indicates how likely the exercise is to be performed based on the recognized exercise duration using the exercise information. The instructions may cause the processor to resume the detection operation based on the determination that the reliability is greater than or equal to the designated reference value. The instructions may cause the processor to maintain the stop of the detection operation based on the determination that the reliability is lower than the reference value.

[0148] The instructions may cause the processor to determine the reliability that indicates how likely the exercise is to be performed based on the recognized exercise duration using the exercise information. The instructions may cause the processor to set the number of steps to start the counting as the default based on the determination that there is no reliability. The instructions may cause the processor to set the number of steps to start the counting as the first value greater than the default based on the determination that the reliability is the low level. The instructions may cause the processor to set the number of steps to start the counting as the second value greater than the first value based on the determination that the reliability is the middle level. The instructions may cause the processor to maintain the stop of the detection operation based on the determination that the reliability is the high level.

[0149] The instructions may cause the processor to determine that there is no reliability based on no recognized exercise duration. The instructions may cause the processor to determine the reliability as the low level based on the fact that the recognized exercise duration is shorter than the rest duration. The instructions may cause the processor to determine the reliability as the middle level based on the fact that the recognized exercise duration is longer than the rest duration and shorter than k times the rest duration. The instructions may cause the processor to determine the reliability as the middle level based on the fact that the recognized exercise duration is k times or longer the rest duration and shorter than N minutes. The instructions may cause the processor to determine the reliability as the high level based on the recognized exercise duration being longer than or equal to k times the rest duration and longer than or equal to N minutes.

[0150] The instructions may cause the processor to receive, as the exercise information, from the wearable device through the wireless communication circuit, the first information including the exercise duration and the rest duration, the second information including the ratio of the exercise duration to the rest duration, or the third information including the multiple value indicating how much longer the exercise duration is compared to the rest duration.

[0151] According to one or more embodiments, a portable electronic device (e.g., the portable electronic device **220**) includes: a touch-sensitive display; an input device; a sensing circuit; a processor connected to the display, the input device, and the sensing circuit; and a memory operatively connected to the processor. The memory may store instructions which cause, when executed, the processor to receive first information indicating that an exercise other than walking and running is being performed, from the display or the input device. The instructions may cause the processor to stop the step detection operation using the sensing circuit based on the reception of the first information. The instructions may cause the processor to receive second information indicating that the exercise is ended, from the display or the input device, while the detection operation is stopped. The instructions may cause the processor to resume the detection operation based on the reception of the second information.

[0152] According to one or more embodiments, a method of operating a portable electronic device (e.g., the portable electronic device **220**) may include receiving, from a wearable device (e.g., wearable device **210**) through a wireless communication circuit of the portable electronic device, first information indicating that an exercise other than walking and running is performing. The method may include stopping a step detection operation using the sensing circuit of the portable electronic device based on the reception of the first information. The method may include receiving second information indicating that the exercise is ended, from the wearable device through the wireless communication circuit, while the detection operation is stopped. The method may include resuming the detection operation based on the reception of the second information.

[0153] The information indicating the first time point along with the first information may be received by the portable electronic device. The method may include subtracting the number of steps counted from the first time point to a second time point at which the first information is received from the accumulated number of steps.

[0154] The method may include receiving the exercise information related to the time when an exercise is performed during a predetermined duration of time from the wearable device through the wireless communication circuit, while the detection operation is stopped. The method may include determining whether to resume the detection operation based on the exercise information.

[0155] The determining of whether to resume the detection operation may include: determining reliability that indicates how likely the exercise is to be performed based on a recognized exercise duration using the exercise information; resuming the detection operation based on the reliability that is determined to be greater than or equal to a designated reference value; and maintaining the stop of the detection operation based on the reliability that is determined to be lower than the reference value.

[0156] The determining of whether to resume the detection operation may further include: determining reliability that indicates how likely the exercise is to be performed based on a recognized exercise duration using the exercise information; setting the number of steps to start counting as a default based on the determination that there is no reliability; setting the number of steps to start the counting as a first value greater than the default based on the determination that the reliability is a low level; setting the number of steps to start the counting as a second value greater than the first value based on the determination that the reliability is a middle level; and maintaining the stop of the detection operation based on the determination that the reliability is a high level.

[0157] The determining of the reliability may include: determining that there is no reliability based on no recognized exercise duration;

[0158] determining the reliability as the low level based on the recognized exercise duration being shorter than a rest duration; determining the reliability as the middle level based on the recognized exercise duration being longer than the rest duration and shorter than k times the rest duration; determining the reliability as the middle level based on the recognized exercise duration being k times or longer the rest duration and shorter than N minutes; and determining the reliability as the

high level based on the recognized exercise duration being k times or longer the rest duration and longer than or equal to N minutes.

[0159] The exercise information may include first information including an exercise duration and a rest duration, second information including a ratio of the exercise duration to the rest duration, or third information including a multiple value that indicates how many times the exercise duration is greater than the rest duration.

[0160] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0161] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0162] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0163] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0164] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only

memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0165] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0166] While various embodiments have been described above with reference to the drawings, the present disclosure is not limited thereto, and any combination or substitution of components as appropriate is included within the scope of the present disclosure. In some embodiments, modifications such as combinations, changes in the order of processes, and various changes in design may be made on the basis of knowledge of a person skilled in the art, and such modified embodiments are within the scope of the present disclosure and the appended claims.

Claims

1. A portable electronic device, comprising: a wireless communication circuit; a sensing circuit; at least one processor connected to the wireless communication circuit and the sensing circuit; and a memory operatively connected to the at least one processor and storing instructions, wherein the at least one processor is configured to execute the instructions to cause the portable electronic device to: perform a step detection operation using the sensing circuit; receive, from a wearable device through the wireless communication circuit, first information indicating that an exercise other than walking and running is being performed; stop the step detection operation using the sensing circuit based on receiving the first information; receive, while the step detection operation is stopped and from the wearable device through the wireless communication circuit, second information indicating that the exercise ended; and resume the step detection operation based on receiving the second information.

2. The portable electronic device of claim 1, wherein the at least one processor is configured to execute the instructions to cause the portable electronic device to: receive a first time point, from the wearable device through the wireless communication circuit, along with the first information; and subtract a number of steps counted from the first time point to a second time point from an accumulated number of steps, the second time point being a time point at which the first information is received by the portable electronic device.

3. The portable electronic device of claim 1, wherein the at least one processor is configured to execute the instructions to cause the portable electronic device to: receive, while the step detection operation is stopped and from the wearable device through the wireless communication circuit, exercise information related to a time when the exercise is performed during a predetermined duration of time; and determine whether to resume the step detection operation based on the exercise information.

4. The portable electronic device of claim 3, wherein the at least one processor is configured to

execute the instructions to cause the portable electronic device to: determine a reliability level indicating how likely the exercise is to be performed based on an exercise duration using the exercise information; resume the step detection operation based on the reliability level being greater than or equal to a reference value; and maintain stopping of the step detection operation based on the reliability level being lower than the reference value.

5. The portable electronic device of claim 3, wherein the at least one processor is configured to execute the instructions to cause the portable electronic device to: determine a reliability level indicating how likely the exercise is to be performed based on an exercise duration using the exercise information; set a number of steps to start counting as a default based on the reliability level being zero; set the number of steps to start the counting as a first value greater than the default based on the determination that the reliability level is a low level; set the number of steps to start the counting as a second value greater than the first value based on the determination that the reliability level is a middle level; and maintain stopping of the step detection operation based on the determination that the reliability level is a high level.

6. The portable electronic device of claim 5, wherein the at least one processor is configured to execute the instructions to cause the portable electronic device to: determine that the reliability level is zero based on the exercise duration being zero; determine the reliability level is the low level based on the exercise duration being shorter than a rest duration; determine the reliability level is the middle level based on the exercise duration being longer than the rest duration and shorter than k times the rest duration or the exercise duration being k times or longer the rest duration and shorter than N minutes; and determine the reliability level is the high level based on the exercise duration being k times or longer the rest duration and longer than or equal to N minutes.

7. The portable electronic device of claim 3, the at least one processor is configured to execute the instructions to cause the portable electronic device to receive, as the exercise information and from the wearable device through the wireless communication circuit, the first information comprising an exercise duration and a rest duration, the second information comprising a ratio of the exercise duration to the rest duration, or third information comprising a multiple value indicating how many times the exercise duration is greater than the rest duration.

8. A portable electronic device, comprising: a touch-sensitive display; a wireless communication circuit; an input device; a sensing circuit; at least one processor connected to the touch-sensitive display, the input device, and the sensing circuit; and a memory operatively connected to the at least one processor and storing instructions, wherein the at least one processor is configured to execute the instructions to cause the portable electronic device to: perform a step detection operation using the sensing circuit; receive, from a wearable device through the wireless communication circuit, first information indicating that an exercise other than walking and running is being performed; stop the step detection operation using the sensing circuit based on receiving the first information; receive, while the step detection operation is stopped and from the wearable device through the wireless communication circuit, second information indicating that the exercise ended; and resume the step detection operation based on receiving the second information.

9. The portable electronic device of claim 8, wherein the at least one processor is configured to execute the instructions to cause the portable electronic device to: receive a first time point, from the wearable device through the wireless communication circuit, along with the first information; and subtract a number of steps counted from the first time point to a second time point from an accumulated number of steps, the second time point being a time point at which the first information is received by the portable electronic device.

10. The portable electronic device of claim 8, wherein the at least one processor is configured to execute the instructions to cause the portable electronic device to: receive, while the step detection operation is stopped and from the wearable device through the wireless communication circuit, exercise information related to a time when the exercise is performed during a predetermined

duration of time; and determine whether to resume the step detection operation based on the exercise information.

11. The portable electronic device of claim 10, wherein the at least one processor is configured to execute the instructions to cause the portable electronic device to: determine a reliability level indicating how likely the exercise is to be performed based on an exercise duration using the exercise information; resume the step detection operation based on the reliability level being greater than or equal to a reference value; and maintain stopping of the step detection operation based on the reliability level being lower than the reference value.

12. The portable electronic device of claim 10, wherein the at least one processor is configured to execute the instructions to cause the portable electronic device to: determine a reliability level indicating how likely the exercise is to be performed based on an exercise duration using the exercise information; set a number of steps to start counting as a default based on the reliability level being zero; set the number of steps to start the counting as a first value greater than the default based on the determination that the reliability level is a low level; set the number of steps to start the counting as a second value greater than the first value based on the determination that the reliability level is a middle level; maintain stopping of the step detection operation based on the determination that the reliability level is a high level; determine that the reliability level is zero based on the exercise duration being zero; determine the reliability level is the low level based on the exercise duration being shorter than a rest duration; determine the reliability level is the middle level based on the exercise duration being longer than the rest duration and shorter than k times the rest duration or the exercise duration being k times or longer the rest duration and shorter than N minutes; and determine the reliability level is the high level based on the exercise duration being k times or longer the rest duration and longer than or equal to N minutes.

13. The portable electronic device of claim 10, the at least one processor is configured to execute the instructions to cause the portable electronic device to receive, as the exercise information and from the wearable device through the wireless communication circuit, the first information comprising an exercise duration and a rest duration, the second information comprising a ratio of the exercise duration to the rest duration, or third information comprising a multiple value indicating how many times the exercise duration is greater than the rest duration.

14. A method of operating a portable electronic device, comprising: performing a step detection operation using a sensing circuit of the portable electronic device; receiving, from a wearable device through a wireless communication circuit of the portable electronic device, first information indicating that an exercise other than walking or running is being performed; stopping the step detection operation using the sensing circuit based on receiving of the first information; receiving, while the step detection operation is stopped and from the wearable device through the wireless communication circuit, second information indicating that the exercise is ended; and resuming the step detection operation based on receiving the second information.

15. The method of claim 14, further comprising: receiving a first time point, from the wearable device through the wireless communication circuit, along with the first information; and subtracting a number of steps counted from the first time point to a second time point from an accumulated number of steps, the second time point being a time point at which the first information is received by the portable electronic device.

16. The method of claim 14, further comprising: receiving, while the step detection operation is stopped and from the wearable device through the wireless communication circuit, exercise information related to a time when the exercise is performed during a predetermined duration of time; and determining whether to resume the step detection operation based on the exercise information.

17. The method of claim 16, wherein the determining of whether to resume the step detection operation comprises: determining a reliability level indicating how likely the exercise is to be performed based on an exercise duration using the exercise information; resuming the step

detection operation based on the reliability level being greater than or equal to a reference value; and maintaining stopping of the step detection operation based on the reliability level being lower than the reference value.

18. The method of claim 16, wherein the determining of whether to resume the step detection operation further comprises: determining a reliability level indicating how likely the exercise is to be performed based on an exercise duration using the exercise information; setting a number of steps to start counting as a default based on the reliability level being zero; setting the number of steps to start the counting as a first value greater than the default based on the determination that the reliability level is a low level; setting the number of steps to start the counting as a second value greater than the first value based on the determination that the reliability level is a middle level; and maintaining stopping of the step detection operation based on the determination that the reliability level is a high level.

19. The method of claim 18, wherein the determining of the reliability level comprises: determining that the reliability level is zero based on the exercise duration being zero; determining the reliability level is the low level based on the exercise duration being shorter than a rest duration; determining the reliability level is the middle level based on the exercise duration being longer than the rest duration and shorter than k times the rest duration or the exercise duration being k times or longer the rest duration and shorter than N minutes; and determining the reliability level is the high level based on the exercise duration being k times or longer the rest duration and longer than or equal to N minutes.

20. The method of claim 16, wherein the exercise information comprises the first information comprising an exercise duration and a rest duration, the second information comprising a ratio of the exercise duration to the rest duration, or third information comprising a multiple value indicating how many times the exercise duration is greater than the rest duration.
